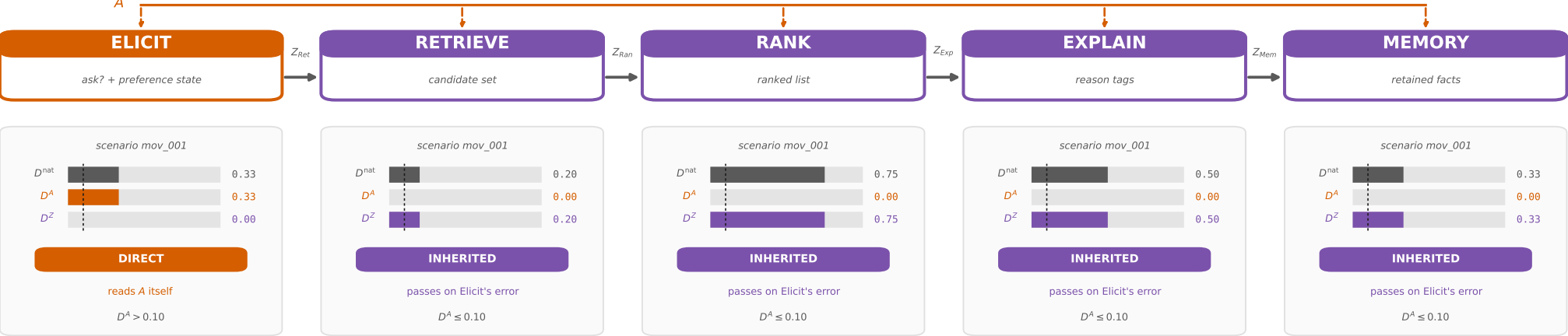


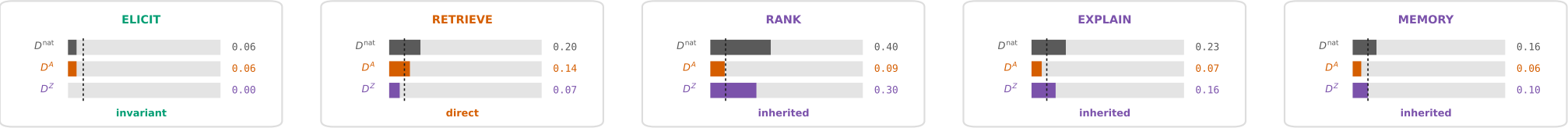
Stage-Crossover Attribution: where protected-attribute dependence enters the pipeline, and where it is only passed along

Each stage is re-run four times -- its own parent state and the other arm's, crossed with each descriptor -- so its divergence splits into a direct and an inherited part

A (protected descriptor) is visible to every stage; Z_s (legitimate parent state) flows left to right



Corpus means over the same design (movies, 24 scenarios x 4 repeats, faults planted uniformly across stages)



What the decomposition buys, measured on this run

Localization	100%	of 48 scenarios: the stage with the largest D^A is the planted stage (or none, when nothing was planted)
Masking	36.5 / 43.8%	of pairs are process-sensitive yet endpoint-equivalent (movies / restaurants) -- invisible to an output-only audit
Repair	0.58 vs 0.17	final-gap removability when the implicated stage is repaired vs. when a wrong stage is